

Music / Playlist Organiser

Working Output

How this output was produced

Every panel in this document is real console output, captured on 4 October 2026 by running the program in this repository against MySQL. Nothing was typed in by hand.

Python	3.14.5
MySQL Server	26.7.0 (with the <code>mysql</code> command-line client for section 2)
Libraries	mysql-connector-python 26.7.0, tabulate 0.10.0
Operating system	Windows 11

The database was freshly created from `music_db.sql` before the run, so IDs match the sample data. Sections 3.3 to 3.11 are one continuous session, so each step builds on the one before. Text typed by the user is **highlighted**.

The output was captured from a copy of the database so that real data was not changed; that copy's name is shown as `music_db`.

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1. Database setup
2. MySQL output: the 12 sample queries
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5. Automated test results (77/77 passed)

1. Database setup

1.1 First run: the program creates the database

If music_db does not exist, the program offers to build it from music_db.sql.

```
The database 'music_db' does not exist yet.
Create it now with the sample songs and playlists? (y/n): y
✅ 'music_db' created.

===== MUSIC / PLAYLIST ORGANISER =====
1. Add Song
2. View Songs
3. Search Song
4. Update Song
5. Delete Song
6. Create Playlist
7. Add Song to Playlist
8. View Playlist
9. Exit
Enter your choice: 9

👋 Exiting program. Goodbye!
```

1.2 Tables

Structure of the three tables, shown by the MySQL command-line client.

```
mysql> SHOW TABLES;
+-----+
| Tables_in_music_db |
+-----+
| playlists           |
| playlistsongs       |
| songs               |
+-----+
3 rows in set (0.024 sec)
```

```
mysql> DESCRIBE Songs;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| song_id    | int           | NO   | PRI | NULL    | auto_increment |
| title      | varchar(100)  | NO   |     | NULL    |                |
| artist     | varchar(100)  | NO   |     | NULL    |                |
| album      | varchar(100)  | YES  |     | NULL    |                |
| genre      | varchar(50)   | YES  |     | NULL    |                |
| duration   | varchar(10)   | YES  |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.042 sec)
```

```
mysql> DESCRIBE Playlists;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| playlist_id    | int           | NO   | PRI | NULL    | auto_increment |
| playlist_name  | varchar(100)  | NO   | UNI | NULL    |                |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.007 sec)
```

```
mysql> DESCRIBE PlaylistSongs;
```

```
+-----+-----+-----+-----+-----+
| Field      | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+
| playlist_id | int  | NO   | PRI | NULL    |       |
| song_id     | int  | NO   | PRI | NULL    |       |
+-----+-----+-----+-----+-----+
2 rows in set (0.019 sec)
```

2. MySQL output: the 12 sample queries

From `sample_queries.sql`, run on the sample data.

1. Display all songs

```
mysql> SELECT * FROM Songs;
```

```
+-----+-----+-----+-----+-----+
| song_id | title                | artist        | album                | genre        | duration |
+-----+-----+-----+-----+-----+
| 1       | Blinding Lights      | The Weeknd    | After Hours          | Pop          | 3:20     |
| 2       | Save Your Tears      | The Weeknd    | After Hours          | Pop          | 3:36     |
| 3       | Starboy              | The Weeknd    | Starboy              | Pop          | 3:50     |
| 4       | Paradise             | Coldplay      | Mylo Xyloto          | Rock         | 4:12     |
| 5       | Hymn for the Weekend | Coldplay      | A Head Full of Dreams | Pop          | 4:18     |
| 6       | Viva La Vida         | Coldplay      | Viva La Vida         | Alternative   | 4:02     |
| 7       | Fix You              | Coldplay      | X&Y                  | Rock         | 4:55     |
| 8       | Shape of You         | Ed Sheeran    | Divide               | Pop          | 3:54     |
| 9       | Perfect              | Ed Sheeran    | Divide               | Pop          | 4:40     |
| 10      | Photograph           | Ed Sheeran    | X                    | Pop          | 4:19     |
| 11      | Thinking Out Loud    | Ed Sheeran    | X                    | Soul         | 4:41     |
| 12      | Believer             | Imagine Dragons | Evolve               | Rock         | 3:24     |
| 13      | Thunder              | Imagine Dragons | Evolve               | Rock         | 3:07     |
| 14      | Radioactive          | Imagine Dragons | Night Visions        | Alternative   | 3:06     |
| 15      | Demons               | Imagine Dragons | Night Visions        | Alternative   | 2:57     |
| 16      | Counting Stars       | OneRepublic   | Native               | Pop Rock     | 4:17     |
| 17      | Apologize            | OneRepublic   | Dreaming Out Loud    | Pop          | 3:28     |
| 18      | Secrets              | OneRepublic   | Waking Up            | Pop          | 3:44     |
| 19      | Levitating           | Dua Lipa      | Future Nostalgia     | Pop          | 3:23     |
| 20      | Don't Start Now      | Dua Lipa      | Future Nostalgia     | Disco        | 3:03     |
| 21      | New Rules            | Dua Lipa      | Dua Lipa             | Pop          | 3:29     |
| 22      | Shivers              | Ed Sheeran    | =                    | Pop          | 3:27     |
| 23      | Stay                 | Justin Bieber | Justice              | Pop          | 2:21     |
| 24      | Peaches              | Justin Bieber | Justice              | R&B          | 3:18     |
| 25      | Sorry               | Justin Bieber | Purpose              | Pop          | 3:20     |
| 26      | Despacito            | Luis Fonsi    | Vida                 | Reggaeton    | 3:47     |
| 27      | Senorita             | Shawn Mendes  | Shawn Mendes         | Pop          | 3:11     |
| 28      | Stitches             | Shawn Mendes  | Handwritten          | Pop          | 3:26     |
| 29      | Attention            | Charlie Puth  | Voicenotes           | Pop          | 3:28     |
| 30      | We Don't Talk Anymore | Charlie Puth  | Nine Track Mind      | Pop          | 3:37     |
+-----+-----+-----+-----+-----+
30 rows in set (0.003 sec)
```

2. Display all playlists

```
mysql> SELECT * FROM Playlists ORDER BY playlist_id;
+-----+-----+
| playlist_id | playlist_name |
+-----+-----+
| 1 | My Favorites |
| 2 | Workout Vibes |
| 3 | Study Chill |
| 4 | Party Hits |
| 5 | Romantic Mood |
| 6 | Road Trip |
+-----+-----+
6 rows in set (0.008 sec)
```

3. Show all songs in "Workout Vibes"

```
mysql> SELECT s.title, s.artist FROM Songs s JOIN PlaylistSongs ps ON s.song_id = ps.song_id WHERE ps.playlist_id = 2;
+-----+-----+
| title | artist |
+-----+-----+
| Paradise | Coldplay |
| Believer | Imagine Dragons |
| Thunder | Imagine Dragons |
| Levitating | Dua Lipa |
| Sorry | Justin Bieber |
+-----+-----+
5 rows in set (0.003 sec)
```

4. Search songs by genre 'Pop'

```
mysql> SELECT title, artist FROM Songs WHERE genre = 'Pop';
+-----+-----+
| title | artist |
+-----+-----+
| Blinding Lights | The Weeknd |
| Save Your Tears | The Weeknd |
| Starboy | The Weeknd |
| Hymn for the Weekend | Coldplay |
| Shape of You | Ed Sheeran |
| Perfect | Ed Sheeran |
| Photograph | Ed Sheeran |
| Apologize | OneRepublic |
| Secrets | OneRepublic |
| Levitating | Dua Lipa |
| New Rules | Dua Lipa |
| Shivers | Ed Sheeran |
| Stay | Justin Bieber |
| Sorry | Justin Bieber |
| Senorita | Shawn Mendes |
| Stitches | Shawn Mendes |
| Attention | Charlie Puth |
| We Don't Talk Anymore | Charlie Puth |
+-----+-----+
18 rows in set (0.003 sec)
```

5. Find all songs by 'Coldplay'

```
mysql> SELECT title, album FROM Songs WHERE artist = 'Coldplay';
```

title	album
Paradise	Mylo Xyloto
Hymn for the Weekend	A Head Full of Dreams
Viva La Vida	Viva La Vida
Fix You	X&Y

```
4 rows in set (0.006 sec)
```

6. Count total songs by each artist

```
mysql> SELECT artist, COUNT(*) AS total_songs FROM Songs GROUP BY artist;
```

artist	total_songs
The Weeknd	3
Coldplay	4
Ed Sheeran	5
Imagine Dragons	4
OneRepublic	3
Dua Lipa	3
Justin Bieber	3
Luis Fonsi	1
Shawn Mendes	2
Charlie Puth	2

```
10 rows in set (0.007 sec)
```

7. List all albums and how many songs they have

```
mysql> SELECT album, COUNT(*) AS total_songs FROM Songs GROUP BY album;
```

album	total_songs
After Hours	2
Starboy	1
Mylo Xyloto	1
A Head Full of Dreams	1
Viva La Vida	1
X&Y	1
Divide	2
X	2
Evolve	2
Night Visions	2
Native	1
Dreaming Out Loud	1
Waking Up	1
Future Nostalgia	2
Dua Lipa	1
=	1
Justice	2
Purpose	1
Vida	1
Shawn Mendes	1
Handwritten	1
Voicenotes	1
Nine Track Mind	1

```
23 rows in set (0.011 sec)
```

8. Find the longest song in the database

```
mysql> SELECT title, artist, duration FROM Songs ORDER BY duration DESC LIMIT 1;
```

title	artist	duration
Fix You	Coldplay	4:55

```
1 row in set (0.007 sec)
```

9. Find songs shorter than 3:30

```
mysql> SELECT title, artist, duration FROM Songs WHERE duration < '3:30';
```

title	artist	duration
Blinding Lights	The Weeknd	3:20
Believer	Imagine Dragons	3:24
Thunder	Imagine Dragons	3:07
Radioactive	Imagine Dragons	3:06
Demons	Imagine Dragons	2:57
Apologize	OneRepublic	3:28
Levitating	Dua Lipa	3:23
Don't Start Now	Dua Lipa	3:03
New Rules	Dua Lipa	3:29
Shivers	Ed Sheeran	3:27
Stay	Justin Bieber	2:21
Peaches	Justin Bieber	3:18
Sorry	Justin Bieber	3:20
Senorita	Shawn Mendes	3:11
Stitches	Shawn Mendes	3:26
Attention	Charlie Puth	3:28

```
16 rows in set (0.007 sec)
```

10. Show songs in 'Romantic Mood' playlist

```
mysql> SELECT s.title, s.artist FROM Songs s JOIN PlaylistSongs ps ON s.song_id = ps.song_id WHERE ps.playlist_id = 5;
```

title	artist
Fix You	Coldplay
Thinking Out Loud	Ed Sheeran
Demons	Imagine Dragons
Stay	Justin Bieber
Senorita	Shawn Mendes

```
5 rows in set (0.005 sec)
```

11. Show all Pop songs with duration > 4 minutes

```
mysql> SELECT title, artist, duration FROM Songs WHERE genre = 'Pop' AND duration > '4:00';
```

title	artist	duration
Hymn for the Weekend	Coldplay	4:18
Perfect	Ed Sheeran	4:40
Photograph	Ed Sheeran	4:19

```
3 rows in set (0.008 sec)
```

12. Display total number of songs in database

```
mysql> SELECT COUNT(*) AS total_songs FROM Songs;
```

total_songs
30

```
1 row in set (0.012 sec)
```

3. Python program output

3.1 Main menu

```
===== MUSIC / PLAYLIST ORGANISER =====
1. Add Song
2. View Songs
3. Search Song
4. Update Song
5. Delete Song
6. Create Playlist
7. Add Song to Playlist
8. View Playlist
9. Exit
Enter your choice:
```

3.2 View all songs (option 2)

Enter your choice: 2

🎵 All Songs:

ID	Title	Artist	Album	Genre	Duration
1	Blinding Lights	The Weeknd	After Hours	Pop	3:20
2	Save Your Tears	The Weeknd	After Hours	Pop	3:36
3	Starboy	The Weeknd	Starboy	Pop	3:50
4	Paradise	Coldplay	Mylo Xyloto	Rock	4:12
5	Hymn for the Weekend	Coldplay	A Head Full of Dreams	Pop	4:18
6	Viva La Vida	Coldplay	Viva La Vida	Alternative	4:02
7	Fix You	Coldplay	X&Y	Rock	4:55
8	Shape of You	Ed Sheeran	Divide	Pop	3:54
9	Perfect	Ed Sheeran	Divide	Pop	4:40
10	Photograph	Ed Sheeran	X	Pop	4:19
11	Thinking Out Loud	Ed Sheeran	X	Soul	4:41
12	Believer	Imagine Dragons	Evolve	Rock	3:24
13	Thunder	Imagine Dragons	Evolve	Rock	3:07
14	Radioactive	Imagine Dragons	Night Visions	Alternative	3:06
15	Demons	Imagine Dragons	Night Visions	Alternative	2:57
16	Counting Stars	OneRepublic	Native	Pop Rock	4:17
17	Apologize	OneRepublic	Dreaming Out Loud	Pop	3:28
18	Secrets	OneRepublic	Waking Up	Pop	3:44
19	Levitating	Dua Lipa	Future Nostalgia	Pop	3:23

20	Don't Start Now	Dua Lipa	Future Nostalgia	Disco	3:03
21	New Rules	Dua Lipa	Dua Lipa	Pop	3:29
22	Shivers	Ed Sheeran	=	Pop	3:27
23	Stay	Justin Bieber	Justice	Pop	2:21
24	Peaches	Justin Bieber	Justice	R&B	3:18
25	Sorry	Justin Bieber	Purpose	Pop	3:20
26	Despacito	Luis Fonsi	Vida	Reggaeton	3:47
27	Senorita	Shawn Mendes	Shawn Mendes	Pop	3:11
28	Stitches	Shawn Mendes	Handwritten	Pop	3:26
29	Attention	Charlie Puth	Voicenotes	Pop	3:28
30	We Don't Talk Anymore	Charlie Puth	Nine Track Mind	Pop	3:37

Total songs: 30

3.3 Add a song (option 1)

```

Enter your choice: 1
Enter Song Title: Is There Someone Else?
Enter Artist: The Weeknd
Enter Album: Dawn FM
Enter Genre: R&B
Enter Duration (mm:ss): 3:19

✅ Song added successfully! (Song ID: 31)

```

3.4 Input checks while adding

An empty title and badly formatted durations are rejected and asked again.

```

Enter your choice: 1
Enter Song Title:
    This field cannot be empty.
Enter Song Title: Hotel
Enter Artist: Montell Fish
Enter Album: Her Love Still Haunts Me Like A Ghost
Enter Genre: R&B
Enter Duration (mm:ss): 3.17
    Please use the format mm:ss, e.g. 3:45
Enter Duration (mm:ss): 3:77
    Please use the format mm:ss, e.g. 3:45
Enter Duration (mm:ss): 3:17

✅ Song added successfully! (Song ID: 32)

```

3.5 Duplicate warning

Adding a song that is already stored (ignoring capital letters) asks first.

```
Enter your choice: 1
Enter Song Title: starboy
Enter Artist: the weeknd

⚠ 'starboy' by the weeknd is already in the database (Song ID: 3).
Add it again anyway? (y/n): n

Song not added.
```

3.6 Search (option 3)

By artist, by genre, and a search with no results.

```
Enter your choice: 3
Enter keyword (title/artist/genre): coldplay
```

🔍 Search Results for 'coldplay':

ID	Title	Artist	Album	Genre	Duration
4	Paradise	Coldplay	Mylo Xyloto	Rock	4:12
5	Hymn for the Weekend	Coldplay	A Head Full of Dreams	Pop	4:18
6	Viva La Vida	Coldplay	Viva La Vida	Alternative	4:02
7	Fix You	Coldplay	X&Y	Rock	4:55

```
Enter your choice: 3
Enter keyword (title/artist/genre): rock
```

🔍 Search Results for 'rock':

ID	Title	Artist	Album	Genre	Duration
4	Paradise	Coldplay	Mylo Xyloto	Rock	4:12
7	Fix You	Coldplay	X&Y	Rock	4:55
12	Believer	Imagine Dragons	Evolve	Rock	3:24
13	Thunder	Imagine Dragons	Evolve	Rock	3:07
16	Counting Stars	OneRepublic	Native	Pop Rock	4:17

```
Enter your choice: 3
Enter keyword (title/artist/genre): taylor swift
```

No songs match 'taylor swift'.

3.7 Update a song (option 4)

Only the genre is changed; pressing Enter keeps the other values. A search confirms the change.

```
Enter your choice: 4
Enter Song ID to update: 31

Current details:

+-----+-----+-----+-----+-----+-----+
| ID   | Title                | Artist   | Album  | Genre  | Duration |
+-----+-----+-----+-----+-----+-----+
| 31   | Is There Someone Else? | The Weeknd | Dawn FM | R&B    | 3:19     |
+-----+-----+-----+-----+-----+-----+
Type a new value, or press Enter to keep the current one.

New Title [Is There Someone Else?]:
New Artist [The Weeknd]:
New Album [Dawn FM]:
New Genre [R&B]: Synth-pop
New Duration (mm:ss) [3:19]:

🔪 Song updated successfully!
```

```
Enter your choice: 3
Enter keyword (title/artist/genre): someone else

🔍 Search Results for 'someone else':

+-----+-----+-----+-----+-----+-----+
| ID   | Title                | Artist   | Album  | Genre  | Duration |
+-----+-----+-----+-----+-----+-----+
| 31   | Is There Someone Else? | The Weeknd | Dawn FM | Synth-pop | 3:19     |
+-----+-----+-----+-----+-----+-----+
```

3.8 Delete a song (option 5)

The song is shown first. Answering n cancels; y deletes it and removes it from its playlist.

```
Enter your choice: 5
Enter Song ID to delete: 25

Song to delete:

+-----+-----+-----+-----+-----+-----+
| ID   | Title  | Artist       | Album  | Genre  | Duration |
+-----+-----+-----+-----+-----+-----+
| 25   | Sorry  | Justin Bieber | Purpose | Pop    | 3:20     |
+-----+-----+-----+-----+-----+-----+
It will also be removed from 1 playlist(s).
Are you sure? (y/n): n

Delete cancelled.
```

Enter your choice: **5**
Enter Song ID to delete: **25**

Song to delete:

ID	Title	Artist	Album	Genre	Duration
25	Sorry	Justin Bieber	Purpose	Pop	3:20

It will also be removed from 1 playlist(s).

Are you sure? (y/n): **y**

 'Sorry' deleted successfully!

3.9 Create a playlist (option 6)

Playlist names must be unique, ignoring capital letters.

Enter your choice: **6**
Enter Playlist Name: **Late Night Drive**

 Playlist 'Late Night Drive' created successfully! (Playlist ID: 7)

Enter your choice: **6**
Enter Playlist Name: **late night drive**

 A playlist named 'late night drive' already exists.

3.10 Add songs to a playlist (option 7)

Songs 32 and 3 were then added to the same playlist in the same way. Adding a song that is already in the playlist is refused.

Enter your choice: **7**

 Playlists:

ID	Playlist	Songs
1	My Favorites	5
2	Workout Vibes	4
3	Study Chill	5
4	Party Hits	5
5	Romantic Mood	5
6	Road Trip	5
7	Late Night Drive	0

Enter Playlist ID: **7**

Enter Song ID (options 2 and 3 show song IDs): **31**

 'Is There Someone Else?' added to 'Late Night Drive' successfully!

Enter your choice: 7

📁 Playlists:

ID	Playlist	Songs
1	My Favorites	5
2	Workout Vibes	4
3	Study Chill	5
4	Party Hits	5
5	Romantic Mood	5
6	Road Trip	5
7	Late Night Drive	3

Enter Playlist ID: 7

Enter Song ID (options 2 and 3 show song IDs): 31

'Is There Someone Else?' is already in 'Late Night Drive'.

3.11 View a playlist (option 8)

Workout Vibes now has 4 songs because song 25 was deleted in 3.8.

Enter your choice: 8

📁 Playlists:

ID	Playlist	Songs
1	My Favorites	5
2	Workout Vibes	4
3	Study Chill	5
4	Party Hits	5
5	Romantic Mood	5
6	Road Trip	5
7	Late Night Drive	3

Enter Playlist ID: 2

📁 Playlist: Workout Vibes

ID	Title	Artist	Album	Genre	Duration
4	Paradise	Coldplay	Mylo Xyloto	Rock	4:12
12	Believer	Imagine Dragons	Evolve	Rock	3:24
13	Thunder	Imagine Dragons	Evolve	Rock	3:07

```

+-----+-----+-----+-----+-----+-----+
| 19   | Levitating | Dua Lipa       | Future Nostalgia | Pop   | 3:23   |
+-----+-----+-----+-----+-----+-----+
4 song(s), total time 14:06

```

Enter your choice: 8

📁 Playlists:

```

+-----+-----+-----+
| ID   | Playlist       | Songs |
+-----+-----+-----+
| 1    | My Favorites   | 5     |
+-----+-----+-----+
| 2    | Workout Vibes  | 4     |
+-----+-----+-----+
| 3    | Study Chill    | 5     |
+-----+-----+-----+
| 4    | Party Hits     | 5     |
+-----+-----+-----+
| 5    | Romantic Mood  | 5     |
+-----+-----+-----+
| 6    | Road Trip      | 5     |
+-----+-----+-----+
| 7    | Late Night Drive | 3     |
+-----+-----+-----+

```

Enter Playlist ID: 7

📁 Playlist: Late Night Drive

```

+-----+-----+-----+-----+-----+-----+
| ID   | Title           | Artist      | Album           | Genre   | Duration |
+-----+-----+-----+-----+-----+-----+
| 3    | Starboy         | The Weeknd  | Starboy         | Pop     | 3:50     |
+-----+-----+-----+-----+-----+-----+
| 31   | Is There Someone Else? | The Weeknd  | Dawn FM        | Synth-pop | 3:19     |
+-----+-----+-----+-----+-----+-----+
| 32   | Hotel           | Montell Fish | Her Love Still Haunts Me Like A Ghost | R&B     | 3:17     |
+-----+-----+-----+-----+-----+-----+
3 song(s), total time 10:26

```

3.12 Handling mistakes

An invalid menu choice, an ID that is not a number, a song that does not exist, and a playlist that does not exist.

Enter your choice: x

❌ Invalid choice, please try again.

Enter your choice: 4

Enter Song ID to update: abc

❌ Please enter a valid ID number.

Enter your choice: 5

Enter Song ID to delete: 999

❌ Song not found!

```
Enter your choice: 8

📁 Playlists:

+-----+-----+-----+
| ID   | Playlist      | Songs |
+-----+-----+-----+
| 1    | My Favorites  | 5     |
+-----+-----+-----+
| 2    | Workout Vibes | 4     |
+-----+-----+-----+
| 3    | Study Chill   | 5     |
+-----+-----+-----+
| 4    | Party Hits    | 5     |
+-----+-----+-----+
| 5    | Romantic Mood | 5     |
+-----+-----+-----+
| 6    | Road Trip     | 5     |
+-----+-----+-----+
| 7    | Late Night Drive | 3    |
+-----+-----+-----+
Enter Playlist ID: 99

❌ Playlist not found!
```

3.13 Exit (option 9)

```
Enter your choice: 9

👋 Exiting program. Goodbye!
```

4. Connection problems

4.1 Wrong MySQL password

The program explains what to fix instead of crashing.

```
❌ MySQL rejected the username/password. Update DB_CONFIG in db_config.py.
```

4.2 MySQL server not running

```
❌ Cannot connect to MySQL on 'localhost'. Make sure MySQL Server is installed and running.
```

5. Automated test results (77/77 passed)

An automated script ran every menu option with valid input, invalid input, boundary values (for example 100 vs 101 characters), SQL injection attempts, emoji and accented text, and connection failures, against MySQL 26.7.0.

Area	Checks	Passed
Startup and connection	13	13
Menu	3	3
Add Song	15	15
View Songs	2	2

Search Song	10	10
Update Song	7	7
Delete Song	6	6
Create Playlist	4	4
Add Song to Playlist	5	5
View Playlist	5	5
Damaged or empty database	3	3
Setup script	4	4
Total	77	77

All checks

ID	What was checked	Result
S01	DB missing + 'n' -> hint, no menu	PASS
S02	DB missing + blank -> treated as no	PASS
S03	DB missing + 'maybe' -> treated as no	PASS
S04	DB missing + 'yes' -> creates DB	PASS
S05	DB missing + 'Y' -> creates DB	PASS
S06	wrong password -> clear message	PASS
S07	server unreachable -> clear message	PASS
S08	setup_database.py wrong password -> message	PASS
S09	server dies mid-session -> message per action, menu survives	PASS
S10	Ctrl+C at menu -> clean exit	PASS
S11	Ctrl+C mid Add Song -> clean exit, nothing saved	PASS
S12	EOF at menu -> clean exit	PASS
S13	EOF mid Update -> clean exit, song unchanged	PASS
M01	7 invalid menu choices each rejected	PASS
M02	choice with spaces ' 2 ' accepted	PASS
M03	9 exits	PASS
A01	valid add -> row stored	PASS
A02	whitespace-only title rejected	PASS
A03	title of exactly 100 chars accepted	PASS
A04	title of 101 chars rejected then re-asked	PASS
A05	empty artist rejected then re-asked	PASS
A06	genre 51 rejected, 50 accepted	PASS
A07	blank album/genre/duration stored as NULL	PASS
A08	12 invalid durations rejected, 999:59 accepted	PASS
A09	0:01 ok, 03:45 -> 3:45, ' 12:05 ' -> 12:05	PASS
A10	100 emoji title + accented/CJK artist round-trip (utf8mb4)	PASS

A11	SQL injection title stored literally, table intact	PASS
A12	title with % and _ stored literally	PASS
A13	leading/trailing spaces trimmed	PASS
A14	duplicate title+artist (any case) warns; 'n' skips	PASS
A15	duplicate warning; 'y' adds anyway	PASS
V01	NULL fields shown blank, never 'None'	PASS
V02	total count line matches DB	PASS
Q01	uppercase keyword matches (case-insensitive)	PASS
Q02	partial artist match	PASS
Q03	genre match	PASS
Q04	no match message	PASS
Q05	empty/blank keyword re-asked	PASS
Q06	'%' searched literally (not a wildcard)	PASS
Q07	'_' searched literally (not a wildcard)	PASS
Q08	injection keyword returns nothing	PASS
Q09	accent-insensitive: 'señorita' finds 'Senorita'	PASS
Q10	101-char keyword rejected	PASS
U01	5 malformed IDs rejected	PASS
U02	huge and missing IDs -> not found	PASS
U03	all-blank update keeps every field	PASS
U04	change only genre	PASS
U05	bad duration re-asked during update	PASS
U06	101-char title rejected during update	PASS
U07	update song with NULL fields (prompt shows [])	PASS
D01	bad / missing ID	PASS
D02	'n', blank, 'no', 'nope' all cancel	PASS
D03	'Y' deletes	PASS
D04	'yes' deletes	PASS
D05	song in 2 playlists -> warns and cascades	PASS
D06	song in no playlist -> no playlist warning	PASS
P01	create playlist	PASS
P02	exact / upper-case / padded duplicates refused	PASS
P03	empty and 101-char names re-asked, 100 accepted	PASS
P04	injection name stored literally	PASS
T01	valid add	PASS
T02	duplicate add refused	PASS
T03	bad / missing playlist ID	PASS

T04	bad / missing song ID	PASS
T05	playlists listed before asking	PASS
W01	Workout Vibes after Believer deleted -> 4 songs	PASS
W02	empty playlist message	PASS
W03	missing / bad playlist ID	PASS
W04	hand-entered bad durations don't crash; only 10:05 counted	PASS
W05	total over an hour	PASS
E01	missing table -> message suggests setup_database.py, other options still work	PASS
E02	setup_database.py 'y' repairs it	PASS
E03	empty tables -> friendly messages	PASS
X01	reset declined -> nothing changed	PASS
X02	reset with 'yes'	PASS
X03	SQL file saved with a UTF-8 BOM still works	PASS
X04	SQL file for a different DB than DB_NAME -> refused, nothing dropped	PASS